Università
di Genova

High Performance Computing

Report Mandelbrot Algorithm

Andrea Cattarinich 5137057

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Contents

1 Architectures

For this project I have used three workstation, my personal one, the workstation provided by the University and Google Colab.

1.1 Personal Workstation

HP Envy 17-ae103nl

CPU	
Model	Intel Core i7-6500U @ 2.50 GHz
Architecture	x86_64
Physical cores	2
Logical threads	4 (Hyper-Threading)
GPU	
Model	NVIDIA GeForce GTX 950M
CUDA Compute Capability	5.0
Driver version	572.16
CUDA Version	12.8
Software and Setup	
Visual Studio	Visual Studio Community 2022 (C++ Desktop workload)
CUDA Toolkit	Installed manually
WSL2	Enabled
NVIDIA Drivers	Version 572.16 (for WSL2)

1.2 University Workstation

Room 210

CPU	
Model	Intel Core i9-12900K (12th Gen)
Architecture	x86_64
CPU(s)	24
Threads per core	2
Cores per socket	16
Socket(s)	1
Max frequency	4.02 GHz
Profiling tools	Intel VTune Profiler, Intel Advisor

To use the Intel oneAPI tools, the environment was loaded with:

```
source /opt/intel/oneapi/setvars.sh
```

1.3 Google Colab

CPU	
Model	Intel(R) Xeon(R) CPU @ 2.20GHz
Architecture	x86_64
Physical cores	1
Logical threads	2 (Hyper-Threading)
Socket(s)	1
GPU	
Model	NVIDIA Tesla T4
CUDA Compute Capability	7.5
Driver version	550.54.15
CUDA Version	12.8

2 Mandelbrot Set

2.1 Definition

The Mandelbrot fractal is defined as a set of points in the complex plane. Each point corresponds to a complex number c.

For every c, we consider the iterative sequence:

$$z_0 = 0, \qquad z_{n+1} = z_n^2 + c$$

We then observe the behavior of z_n as $n \to \infty$.

2.1.1 Point that belongs to the Mandelbrot set

Take $c = -1$. Then the sequence is

$$z_0 = 0, \quad z_1 = 0^2 + (-1) = -1, \quad z_2 = (-1)^2 + (-1) = 0, \quad z_3 = 0^2 + (-1) = -1, \ldots$$

The results remain bounded, so $c = -1$ belongs to the Mandelbrot set.

2.1.2 Point that does NOT belong to the Mandelbrot set

Take $c = 1$. Then the sequence is

$$z_0 = 0, \quad z_1 = 0^2 + 1 = 1, \quad z_2 = 1^2 + 1 = 2, \quad z_3 = 2^2 + 1 = 5, \ldots$$

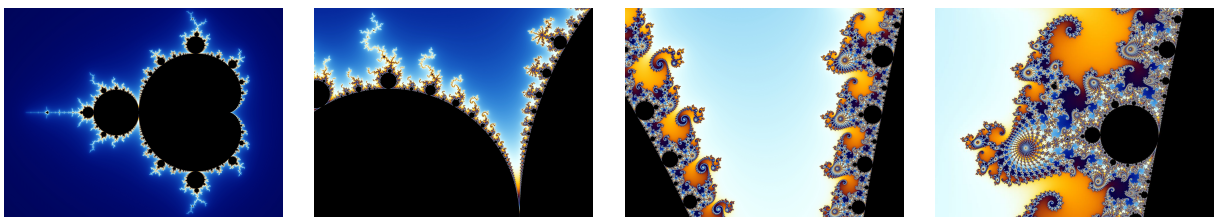
The values grow up to infinite, so $c = 1$ does not belong to the Mandelbrot set.

In our specific case, the escape condition is defined as:

$$|z_i| > 2$$

2.2 Images

This process gives rise to the famously intricate structure of the Mandelbrot fractal. Here a simple explanation [1] or here a more complex one [2] and here an online demo [3].



The color represents how quickly the values of z_n grow:

Black - the sequence remains bounded;

Other colors - the sequence escapes to infinity, with the color indicating how fast it diverges.

3 Hotspot Analysis

3.1 Optimization Problems

Questa istruzione non e' vettorizzabile con MSVC, e per tanto, l'esecuzione risulta essere poco performante.

Mentre con Linux G++, l'operazione sembra ottimizzare `std::complex`.

```
    z = pow(z, 2) + c;
```

L'operazione `std::complex` utilizza l'overloading, in particolare `operator+` e `operator*`, creando un overhead e un conseguente carico sulla CPU. Inoltre, l'operazione `abs(z)` calcola la radice quadrata ogni volta.

Per rimuovere questa dipendenza dovuta dai numeri complessi, ho realizzato un miglioramento nel codice che mi consente di usare solamente numeri reali. Infatti, sfruttando la seguente relazione

$$z^2 = (a, b)^2 = (a^2 - b^2,\ 2ab)$$

Ho implementato:

$$z^2 = \big(\mathrm{Re}(z), \mathrm{Im}(z)\big)^2 = \big(\mathrm{Re}(z)^2 - \mathrm{Im}(z)^2,\ 2\ \mathrm{Re}(z)\,\mathrm{Im}(z)\big)$$

Come segue:

```
    double zRe = 0, zIm = 0;
    for (int i = 1; i <= ITERATIONS; i++)
    {
        double oldRe = zRe;
        double zRe2 = zRe*zRe; // Re(z)^2
        double zIm2 = zIm*zIm; // Im(z)^2

        zRe = (zRe2 - zIm2) + cRe;
        zIm = 2*oldRe*zIm + cIm;

        if(zRe2 + zIm2 >= 4)
        {
            image[pos] = i;
            break;
        }
    }
```

3.1.1 Solution and Results

I risultati, come ci aspettavamo sono più performanti.

MSVC

```
cl /O2 /arch:AVX2 /fp:fast /EHsc src\*.cpp /Fe:build\*.exe
```

WSL

```
g++ -O2 -o build/* src/*.cpp
```

	mandelbrot	opt1	opt2
WSL	96s	30s (x3.2)	36s (x2.7)
Windows	-	40s	36 s

Utilizzeremo questo codice come baseline per lo sviluppo della restante parte del progetto.

3.2 Hotspot Identification

The measurements were performed using Intel VTune Profiler and the analysis was performed using Intel Advisor.

Function Stack	Total	Percentage (%)
Total	69.97	100.00
_start	69.97	100.00
__libc_start_main_impl	69.97	100.00
– main	69.97	100.00
— std::abs<double>	41.2121	58.90
— std::operator+<double>	13.6361	19.49
— std::pow<double>	10.712	15.31
— std::complex<double>::complex	0.0199997	0.03

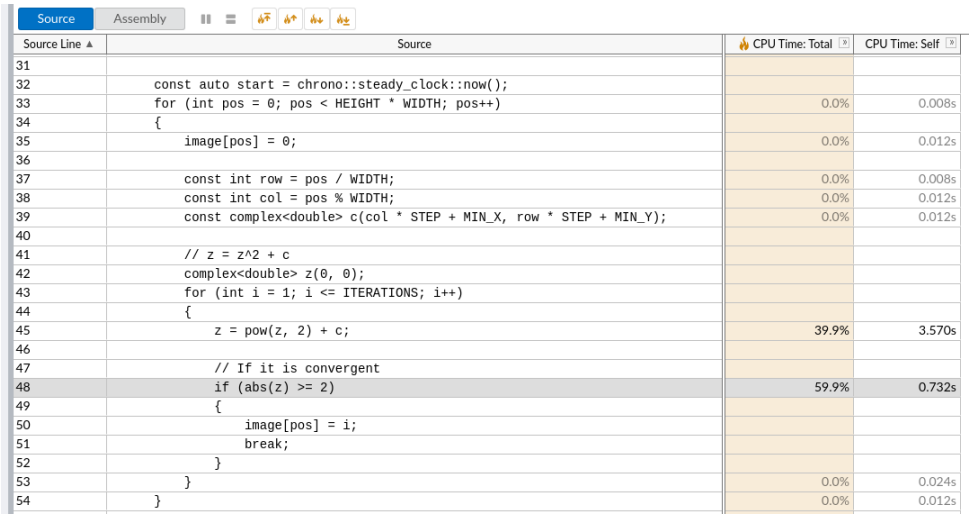
The two main hotspots are the following instructions which takes the most fraction of the program execution.

Line 48

```
1     z = pow(z, 2) + c;
```

Line 51

```
1     if (abs(z) >= 2)
```



Source | Assembly

Source Line	Source	CPU Time: Total	CPU Time: Self
31			
32	const auto start = chrono::steady_clock::now();		
33	for (int pos = 0; pos < HEIGHT * WIDTH; pos++)	0.0%	0.008s
34	{		
35	image[pos] = 0;	0.0%	0.012s
36			
37	const int row = pos / WIDTH;	0.0%	0.008s
38	const int col = pos % WIDTH;	0.0%	0.012s
39	const complex<double> c(col * STEP + MIN_X, row * STEP + MIN_Y);	0.0%	0.012s
40			
41	// z = z^2 + c		
42	complex<double> z(0, 0);		
43	for (int i = 1; i <= ITERATIONS; i++)		
44	{		
45	z = pow(z, 2) + c;	39.9%	3.570s
46			
47	// If it is convergent		
48	if (abs(z) >= 2)	59.9%	0.732s
49	{		
50	image[pos] = i;		
51	break;		
52	}		
53	}	0.0%	0.024s
54	}	0.0%	0.012s

3.3 Conclusions

The main hotspot is the DFT() function and its inverse IDFT(), which together account for over 99,05% of the total execution time. All other functions have negligible impact. This indicates that **parallelization and vectorization efforts should focus on the double loop inside DFT()**.

The main hotspot is the loop contained in lines 72-80 (source: `src/omp_homework.c`). The dominant operations contributing to the algorithm's computational intensity are the `sin()` and `cos()` trigonometric functions, which account for the majority of the execution time, as shown in the following code.

Source | Assembly

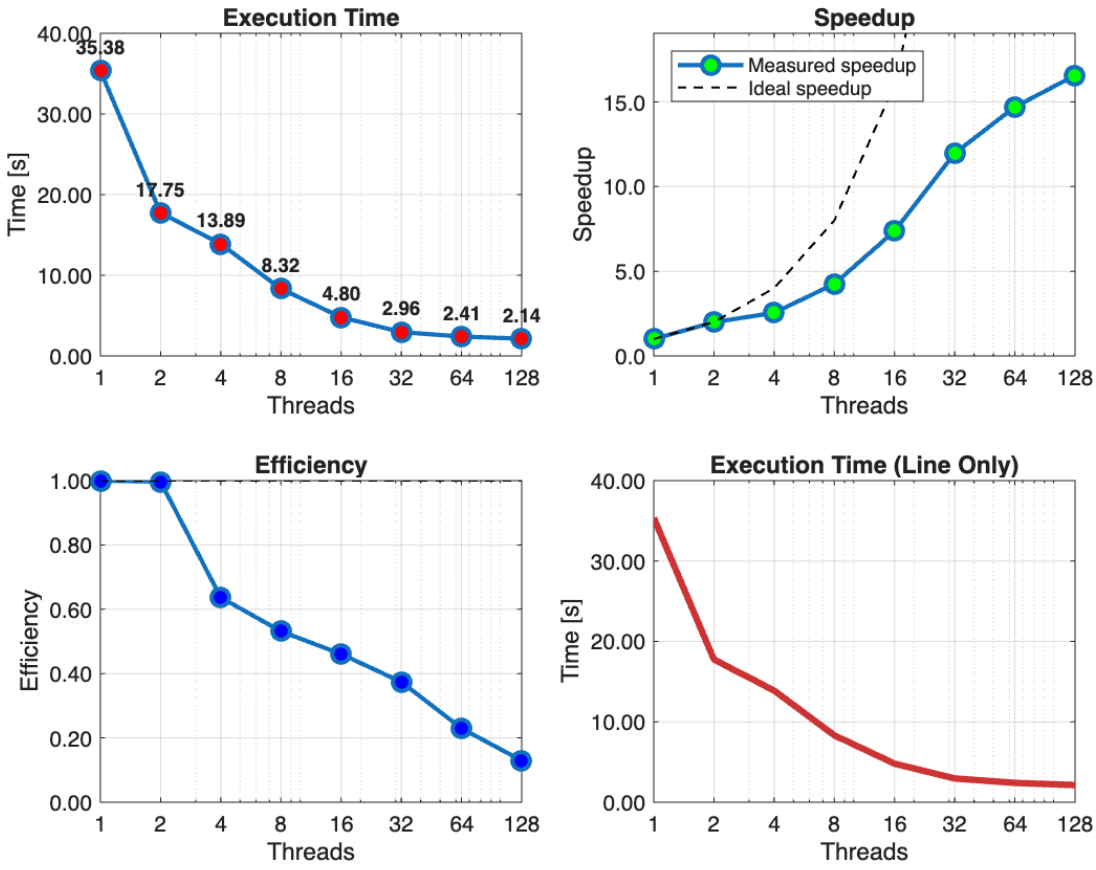
Source Line ▲	Source	CPU Time: Total	CPU Time: Self
31			
32	`const auto start = chrono::steady_clock::now();`		
33	`for (int pos = 0; pos < HEIGHT * WIDTH; pos++)`	0.0%	0.008s
34	`{`		
35	`image[pos] = 0;`	0.0%	0.012s
36			
37	`const int row = pos / WIDTH;`	0.0%	0.008s
38	`const int col = pos % WIDTH;`	0.0%	0.012s
39	`const complex<double> c(col * STEP + MIN_X, row * STEP + MIN_Y);`	0.0%	0.012s
40			
41	`// z = z^2 + c`		
42	`complex<double> z(0, 0);`		
43	`for (int i = 1; i <= ITERATIONS; i++)`		
44	`{`		
45	`z = pow(z, 2) + c;`	39.9%	3.570s
46			
47	`// If it is convergent`		
48	`if (abs(z) >= 2)`	59.9%	0.732s
49	`{`		
50	`image[pos] = i;`		
51	`break;`		
52	`}`		
53	`}`	0.0%	0.024s
54	`}`	0.0%	0.012s

4 OpenMP

4.1 Parallel For

In the first approach (`src/omp1.cpp`), the outer loop over was parallelized by the directive using default static scheduling (`schedule(static)`). Evenly divides pixels among available threads.

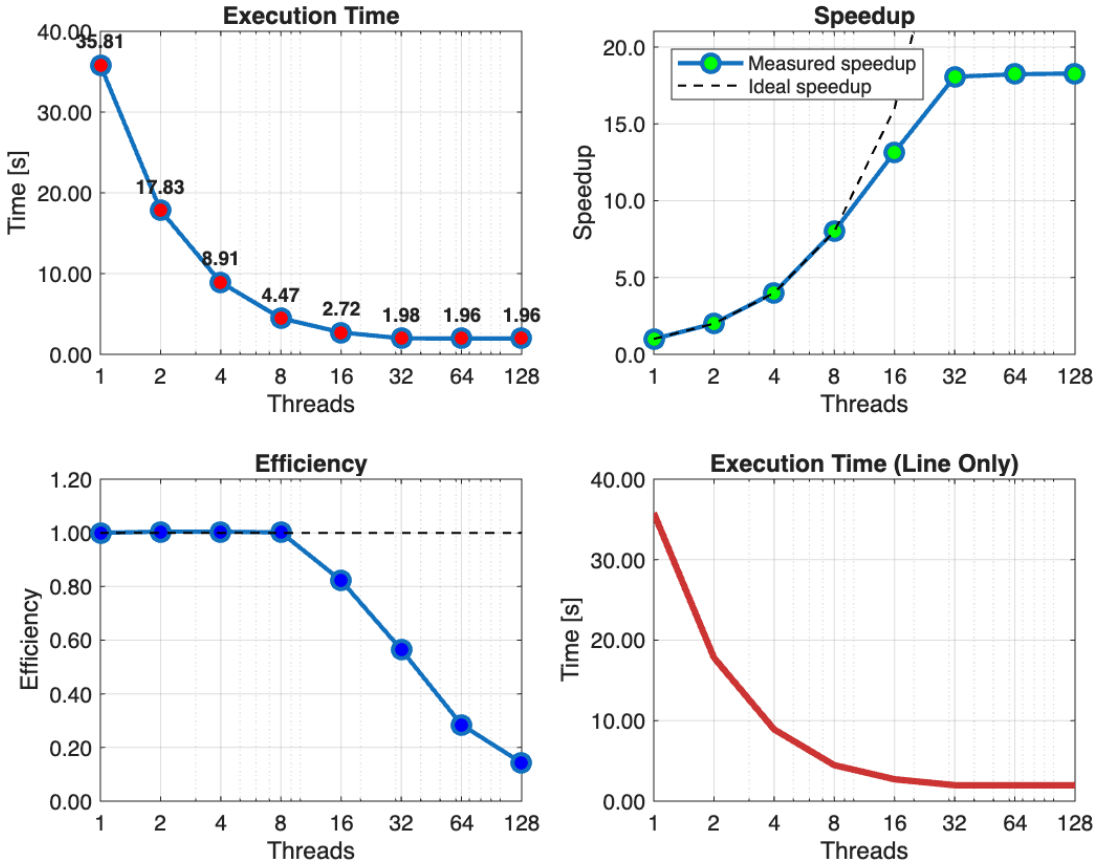
```
    #pragma omp parallel for
    for (int pos = 0; pos < HEIGHT * WIDTH; pos++)
    {
        image[pos] = 0;

        const int row = pos / WIDTH;
        const int col = pos % WIDTH;

        double cRe = col * STEP + MIN_X;
        double cIm = row * STEP + MIN_Y;

        // z = z^2 + c
        double zRe = 0, zIm = 0;
        for (int i = 1; i <= ITERATIONS; i++)
        {
            double oldRe = zRe;
            zRe = (zRe*zRe - zIm*zIm) + cRe;
            zIm = 2*oldRe*zIm + cIm;

            // If it is convergent
            if(zRe*zRe + zIm*zIm >= 4)
            {
                image[pos] = i;
                break;
            }
        }
    }
```



4.2 Guided Scheduling

The following scheduling policy dynamically balances workload distribution. It is essential for handling the computational irregularity of the Mandelbrot set algorithm because different pixels requires different amount of iterations.

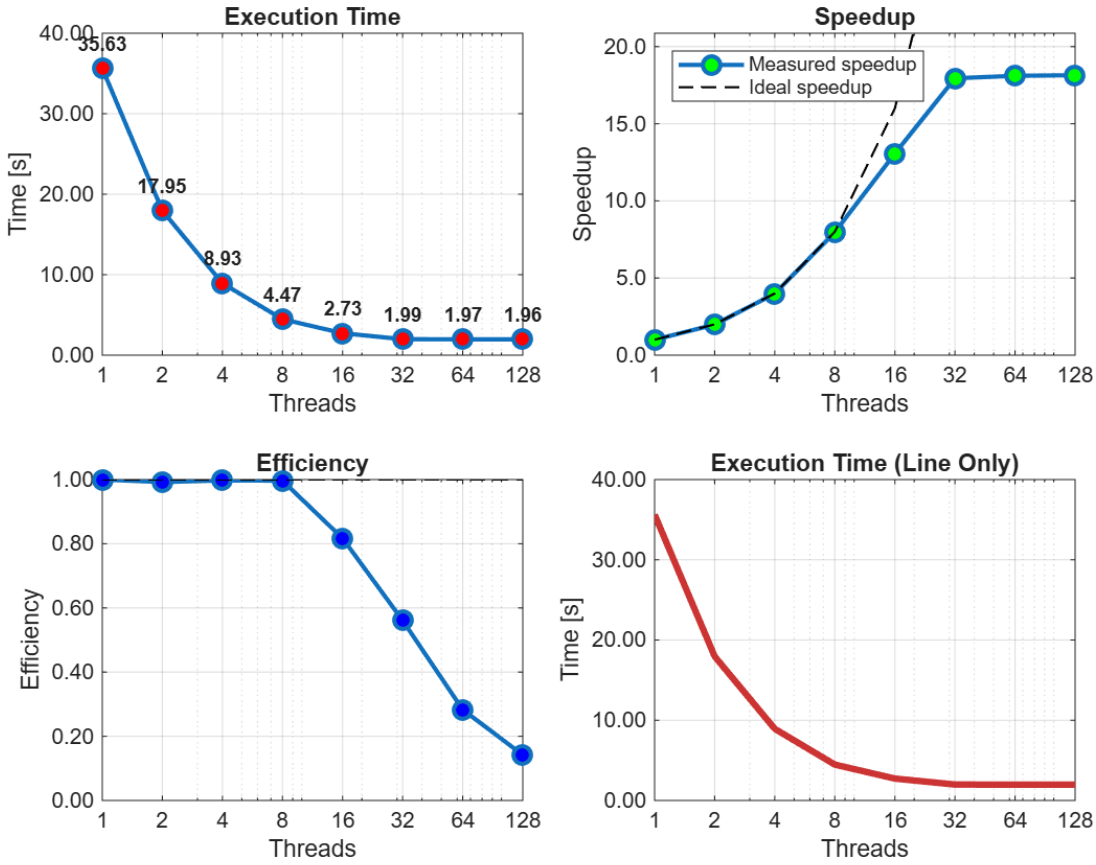
```
    #pragma omp parallel schedule(guided)
    for (int pos = 0; pos < HEIGHT * WIDTH; pos++)
    {
        // ... same content ...
    }
```



4.3 Guided + SIMD

Maximum optimization combining multi-thread parallelism with SIMD vectorization.

```
    #pragma omp parallel for simd schedule(guided)
    for (int pos = 0; pos < HEIGHT * WIDTH; pos++)
    {
        // ... same content ...
    }
```



Here, each thread computes a subset of the output frequencies independently. The temporary variables `Xr_temp` and `Xi_temp` are private to each thread, ensuring there is no race condition when updating the output arrays.

In the second approach, the inner loop over the time index n is parallelized using a reduction clause:

```
for (k = 0; k < N; k++) {
    double Xr_temp = 0.0;
    double Xi_temp = 0.0;
#pragma omp parallel for reduction(+:Xr_temp, Xi_temp)
    for (n = 0; n < N; n++) {
        Xr_temp += xr[n] * cos(n * k * PI2 / N) + idft * xi[n] *
           sin(n * k * PI2 / N);
        Xi_temp += -idft * xr[n] * sin(n * k * PI2 / N) + xi[n] *
            cos(n * k * PI2 / N);
    }
    Xr_o[k] = Xr_temp;
    Xi_o[k] = Xi_temp;
}
```

Here, the computation of each output frequency $X[k]$ is parallelized across the time samples n. The `reduction` clause ensures that each thread accumulates contributions to `Xr_temp` and `Xi_temp` safely, avoiding race conditions. This method can provide higher parallel efficiency for large N, but incurs some overhead from the reduction operation, as shown in the following figures.

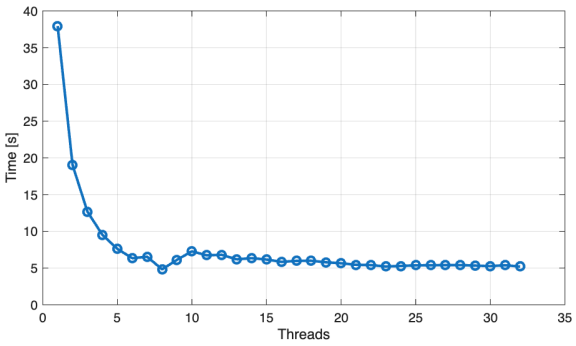
Parallelized `for` loop with OpenMP using `private(n)` to make each thread independent.

```
#pragma omp parallel for private(n)
for (k=0 ; k<N ; k++)
{
    double Xr_temp = 0.0;
    double Xi_temp = 0.0;

    for (n=0 ; n<N ; n++)  {
        Xr_temp += ...
        Xi_temp += ...
    }

    Xr_o[k] = Xr_temp;
    Xi_o[k] = Xi_temp;
}
```

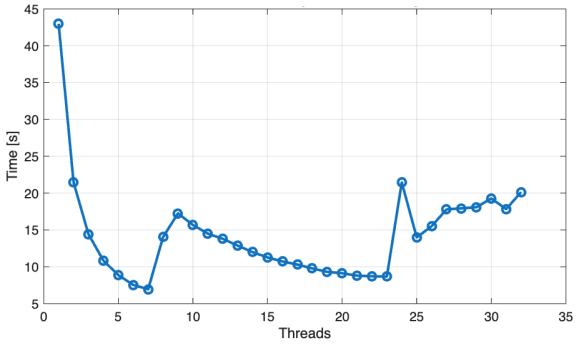
Parallelized `for` loop with OpenMP using `reduction` to combine the results from all threads.

```
int k, n;
for (k=0 ; k<N ; k++)
{
    double Xr_temp = 0.0;
    double Xi_temp = 0.0;

#pragma omp parallel for reduction(+:
    Xr_temp, Xi_temp)
    for (n=0 ; n<N ; n++)  {
        Xr_temp += ...
        Xi_temp += ...
    }

    Xr_o[k] = Xr_temp;
    Xi_o[k] = Xi_temp;
}
```

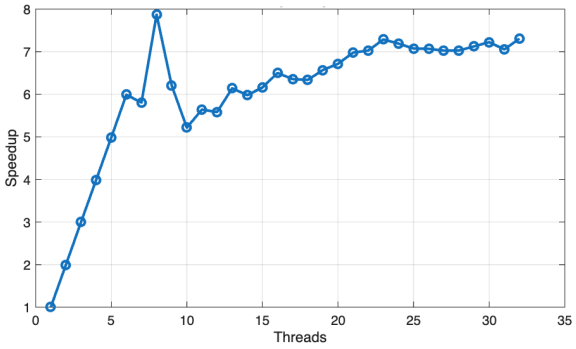
Execution time



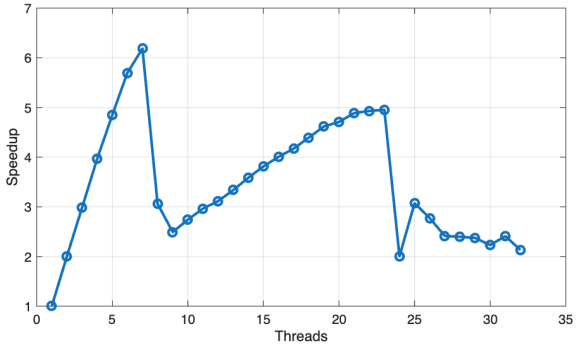
Execution time



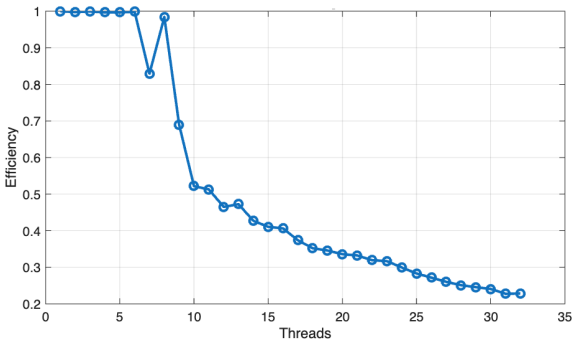
Speedup



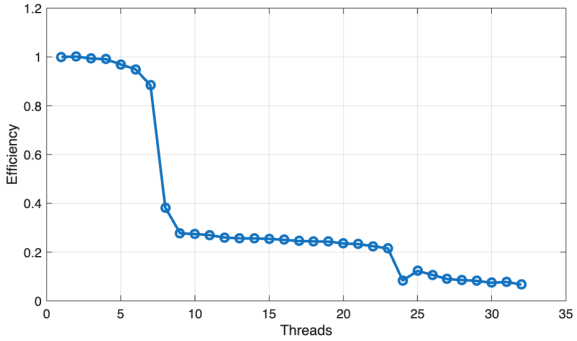
Speedup



Efficiency



Efficiency



Precomputation of Trigonometric Functions: in an additional attempt to optimize the DFT computation, I modified the code as in the source `src/omp3.c`. The idea was to reduce the repeated evaluation of trigonometric functions by computing the sine and cosine values once per iteration:

```
#pragma omp parallel for private(n)
for (k=0; k<N; k++) {
    double Xr_temp = 0.0;
    double Xi_temp = 0.0;

#pragma omp simd reduction(+:Xr_temp, Xi_temp)
    for (n=0; n<N; n++) {
        double angle = n * k * PI2 / N;
        double c = cos(angle);
        double s = sin(angle);

        Xr_temp += xr[n]*c + idft*xi[n]*s;
        Xi_temp += -idft*xr[n]*s + xi[n]*c;
    }

    Xr_o[k] = Xr_temp;
    Xi_o[k] = Xi_temp;
}
```

The motivation behind this modification was that, theoretically, computing the trigonometric functions only once per iteration should reduce the overall workload and improve performance.

However, the experiments did not yield significant improvements. Performance remained the same, or in some cases slightly worse. This is likely due to the compiler automatically applying this optimization by default.

5 CUDA

5.1 Sequential section

Ho deciso inizialmente di eseguire i test sul mio PC personale, piuttosto che sulla workstation dell'università o su Google Colab, per evitare limitazioni dovute a sessioni temporanee e risorse limitate. In questo modo ho avuto maggiore controllo sul processo di compilazione e sull'esecuzione dei test.

Questa parte del report fa riferimento solo all'analisi eseguita tramite il mio PC personale. I parametri della simulazione utilizzati sono:

- $nstep = 200$
- $ni = 15000$
- $nj = 15000$
- $size = ni \times nj = 900\,000\,000$

Ho compilato il codice sequenziale presente in `src/cuda.c` using the following options:

```
gcc -O3 src/cuda.c -o build/cuda
```

Il tempo di esecuzione migliore ottenuto ($37s$) costituirà il riferimento sequenziale per il calcolo dello speedup nella versione GPU.

5.2 CUDA

5.2.1 Code

La parallelizzazione è stata realizzata utilizzando CUDA per eseguire il calcolo dei passi temporali su GPU. Il kernel principale è `step_kernel_mod`, che replica il comportamento della funzione sequenziale `step_kernel_ref`.

Il codice si può trovare in `src/cuda.cu`

Struttura del kernel:

- Ogni thread gestisce un punto della griglia 2D.
- Gli indici globali i e j vengono calcolati a partire dai `threadIdx` e `blockIdx`.

Gestione della memoria:

- Allocazione della memoria host e device.
- Copia dei dati iniziali dal host alla device.
- Copia dei risultati dalla device al host al termine della simulazione.

Controllo errori:

- Dopo ogni chiamata al kernel viene eseguito `cudaGetLastError()`.
- Viene calcolato l'errore massimo rispetto al riferimento CPU per verificare la correttezza.

```
__global__ void step_kernel_mod(int ni, int nj, float fact, float
    * temp_in, float* temp_out)
{
  int i00, im10, ip10, i0m1, i0p1;
  float d2tdx2, d2tdy2;

  int i = threadIdx.x + blockIdx.x * blockDim.x;
  int j = threadIdx.y + blockIdx.y * blockDim.y;

  if(i > 0 && i < ni-1 && j > 0 && j < nj-1){
    // find indices into linear memory
    // for central point and neighbours
    i00 = I2D(ni, i, j);
    im10 = I2D(ni, i-1, j);
    ip10 = I2D(ni, i+1, j);
    i0m1 = I2D(ni, i, j-1);
    i0p1 = I2D(ni, i, j+1);

    // evaluate derivatives
    d2tdx2 = temp_in[im10]-2*temp_in[i00]+temp_in[ip10];
    d2tdy2 = temp_in[i0m1]-2*temp_in[i00]+temp_in[i0p1];

    // update temperatures
    temp_out[i00] = temp_in[i00]+fact*(d2tdx2 + d2tdy2);
  }
}
```

5.2.2 Compilation

La compilazione CUDA è stata effettuata con il comando:

```
nvcc -O3 -arch=sm_50 cuda.cu -o cuda.exe -DDEBUG=0
```

Significato dei flag:

- `-O3`: ottimizzazione massima.
- `-arch=sm_50`: architettura target della GPU.
- `-DDEBUG=0`: disabilita il debug per migliorare le performance.

5.2.3 Execution

Per il kernel CUDA, la scelta di `threadsPerBlock` e `numBlocks` è stata la seguente:

```
    dim3 threadsPerBlock(16, 16);
    dim3 numBlocks((ni + 15) / 16, (nj + 15) / 16);

    // ---- CUDA events start ----
    cudaEvent_t start, stop;
    cudaEventCreate(&start);
    cudaEventCreate(&stop);

    // Execute the modified version using same data
    printf("Execute GPU kernel...\n");
    cudaEventRecord(start, 0); // start timing

    for (istep = 0; istep < nstep; istep++) {
        step_kernel_mod <<<numBlocks, threadsPerBlock>>> (ni, nj,
            tfac, d_temp1, d_temp2);
        cudaDeviceSynchronize();

        cudaError_t err = cudaGetLastError();
        if (err != cudaSuccess)
            printf("CUDA Error: %s\n", cudaGetErrorString(err));

        float* d_tmp = d_temp1;
        d_temp1 = d_temp2;
        d_temp2 = d_tmp;
    }

    cudaEventRecord(stop, 0); // stop timing
    cudaEventSynchronize(stop);
```

Ogni blocco di 16x16 thread calcola i punti della griglia, mentre i blocchi sono organizzati per coprire l'intera matrice.
La temporizzazione è stata eseguita tramite `cudaEvent_t` per misurare il tempo del kernel GPU.

5.2.4 Results

Profiling (command: `nvprof build/cuda`)

950M:

```
Execute the CPU-only reference version...
- Elapsed time: 0.188 s
Execute GPU kernel...
- Elapsed time: 140.515 ms
Max Error = 0.000008
The Max Error of 0.00001 is within ACCEPTABLE bounds.
- Total elapsed time: 2.460 s
==768== Profiling application: build/cuda
==768== Profiling result:
Type  Time(%)      Time     Calls       Avg       Min       Max  Name
GPU:   72.95%  202.16ms         2  101.08ms  2.7665ms  199.40ms  [CUDA memcpy HtoD]
       24.93%  69.101ms       200  345.50us  342.59us  355.94us  step_kernel_mod(int, int, float, float*, float*)
        2.12%  5.8774ms         2  2.9387ms  2.5962ms  3.2812ms  [CUDA memcpy DtoH]
```

```
API:   90.99%  1.69655s         2  848.27ms  699.40us  1.69585s  cudaMalloc
        7.10%  132.39ms       200  661.97us  423.90us  7.6760ms  cudaDeviceSynchronize
        0.96%  17.932ms         4  4.4831ms  3.1292ms  7.7131ms  cudaMemcpy
        0.45%  8.4117ms         2  4.2059ms  2.4000us  8.4093ms  cudaEventCreate
        0.43%  8.0093ms       200  40.046us  9.9000us  1.0583ms  cudaLaunchKernel
        0.04%  796.50us         2  398.25us  302.30us  494.20us  cudaFree
        0.01%  213.70us       200  1.0680us     200ns  31.300us  cudaGetLastError
        0.00%  90.700us         1  90.700us  90.700us  90.700us  cudaEventSynchronize
        0.00%  81.700us       101     808ns     200ns  31.300us  cuDeviceGetAttribute
        0.00%  30.100us         2  15.050us  10.700us  19.400us  cudaEventRecord
        0.00%  5.7000us         2  2.8500us  1.3000us  4.4000us  cudaEventDestroy
        0.00%  4.3000us         1  4.3000us  4.3000us  4.3000us  cuDeviceGetPCIBusId
        0.00%  3.9000us         1  3.9000us  3.9000us  3.9000us  cudaEventElapsedTime
        0.00%  2.3000us         3     766ns     400ns  1.4000us  cuDeviceGetCount
        0.00%  1.6000us         2     800ns     300ns  1.3000us  cuDeviceGet
        0.00%  1.5000us         1  1.5000us  1.5000us  1.5000us  cuDeviceGetName
        0.00%     800ns         1     800ns     800ns     800ns  cuDeviceTotalMem
        0.00%     400ns         1     400ns     400ns     400ns  cuDeviceGetUuid
```

T4:

```
Execute the CPU-only reference version...
- Elapsed time: 36.761 s
Execute GPU kernel...
- Elapsed time: 1741.090 ms
Max Error = 0.000015
The Max Error of 0.00002 is within ACCEPTABLE bounds.
- Total elapsed time: 46.074 s
==5152== Profiling application: /content/drive/MyDrive/Colab Notebooks/HPC-CUDA/build/cuda
==5152== Profiling result:
Type  Time(%)      Time     Calls       Avg       Min       Max  Name
GPU:   67.30%  1.73103s       200  8.6551ms  8.0181ms  11.542ms  step_kernel_mod(int, int, float, float*, float*)
       16.79%  431.89ms         2  215.95ms  203.45ms  228.45ms  [CUDA memcpy HtoD]
       15.91%  409.13ms         2  204.56ms  201.12ms  208.01ms  [CUDA memcpy DtoH]
API:   62.34%  1.73679s       200  8.6840ms  8.0240ms  11.548ms  cudaDeviceSynchronize
       30.23%  842.24ms         4  210.56ms  201.46ms  228.69ms  cudaMemcpy
        7.16%  199.38ms         2  99.688ms  144.08us  199.23ms  cudaMalloc
        0.13%  3.7350ms       200  18.674us  5.4010us  429.51us  cudaLaunchKernel
        0.13%  3.6890ms         2  1.8445ms  870.86us  2.8181ms  cudaFree
        0.01%  168.47us       114  1.4770us     185ns  65.476us  cuDeviceGetAttribute
        0.00%  67.106us       200     335ns     105ns     674ns  cudaGetLastError
        0.00%  33.409us         2  16.704us  10.078us  23.331us  cudaEventRecord
        0.00%  27.898us         2  13.949us  1.1280us  26.770us  cudaEventCreate
        0.00%  12.296us         1  12.296us  12.296us  12.296us  cuDeviceGetName
        0.00%  6.8260us         1  6.8260us  6.8260us  6.8260us  cuDeviceGetPCIBusId
        0.00%  4.4120us         1  4.4120us  4.4120us  4.4120us  cudaEventSynchronize
        0.00%  3.6190us         1  3.6190us  3.6190us  3.6190us  cudaEventElapsedTime
        0.00%  3.3080us         2  1.6540us  1.1110us  2.1970us  cudaEventDestroy
        0.00%  1.6790us         3     559ns     233ns  1.2100us  cuDeviceGetCount
        0.00%     716ns         2     358ns     179ns     537ns  cuDeviceGet
        0.00%     529ns         1     529ns     529ns     529ns  cuDeviceTotalMem
        0.00%     372ns         1     372ns     372ns     372ns  cuModuleGetLoadingMode
        0.00%     302ns         1     302ns     302ns     302ns  cuDeviceGetUuid
```

I risultati della simulazione GPU sono stati confrontati con la versione CPU e lo speedup è calcolato come:

$$\text{speedup} = \frac{T_{\text{CPU}}}{T_{\text{GPU}}}$$

GPU	$\mathbf{T_{CPU}}$ (ms)	$\mathbf{T_{GPU}}$ (ms)	**Speedup**
950M	34967	15086	x2.32
T4	37178	1645	x22.60

Table 1: Confronto tra tempi CPU e GPU e speedup

6 Conclusions

References

[1] Simple Explanation of the Mandelbrot Set.
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https://math.hws.edu/eck/js/mandelbrot/MB.html